
Project Weekly Report 2

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1 Weekly Progress Overview

Last week, our team replicated the foundational concepts presented in the ICLR 2023 paper, "Diffusion Policies as an Expressive Policy Class for Offline Reinforcement Learning" by Wang et al.

Inspired by the article "Efficient Diffusion Policies for Offline Reinforcement Learning", we improved the algorithm in the toy example this week, aiming to speed up the training. This new paper presents the Efficient Diffusion Policy (EDP) to tackle two major drawbacks of Diffusion-QL: (1) High computational costs from full Markov chain propagation during training. (2) Incompatibility with likelihood-based RL algorithms due to diffusion models' intractable likelihood.

2 Brief Introduction to EDP

Construction of EDP

EDP is based on the denoising diffusion probabilistic model (DDPM), which learns a noise-prediction network to predict the noise used to corrupt an example.

In the forward diffusion process, a corrupted sample follows a predefined Gaussian distribution when the clean example and timestep are given. In turn, given a corrupted sample and predicted noise, EDP can approximate its clean version by leveraging the reparametrization trick. Based on this observation, to avoid the tedious sampling process, EDP uses a technique called action approximation to build an action from a corrupted one, which can be easily constructed from the dataset. In this way, each training step only needs to pass through the noise-prediction network once, thus substantially reducing the training time.

EDP employs a DPM-Solver to accelerate the training, which can reduce the sampling steps from 1000 to 15 compared to diffusion-QL using DDPM sampling.

As experimented, simply adding action approximation will obtain **2x** speed-up without performance loss.

Energy-based Action Selection (EAS)

The goal of (offline) RL is to learn a policy that can maximize the cumulative return or values. Though the policy π_θ is stochastic, the learned Q_φ provides a deterministic critic for action evaluation. EAS first samples N actions from π_θ by using any samplers (i.e., DPM-Solver), then samples one of them with weights proportional to $e^{Q(s,a)}$, which eliminates randomness.

This procedure can be understood as sampling from an improved policy:

$$p(a|s) \propto e^{Q(s,a)} \pi_\theta(a|s) \quad (1)$$

3 Detailed Experimental Work

The experimental setup mirrors that of Report 1. We evaluated EDP on the same simple bandit tasks.

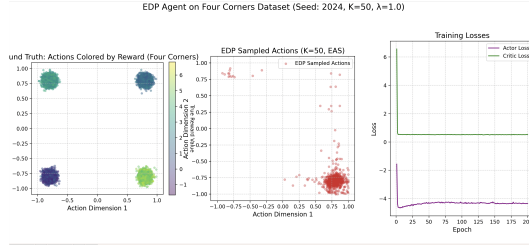


Figure 1: Equal Gaussian Mixture Distribution

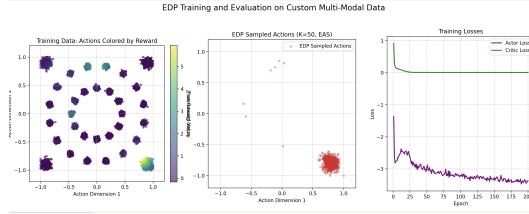


Figure 2: Increased Modal Complexity

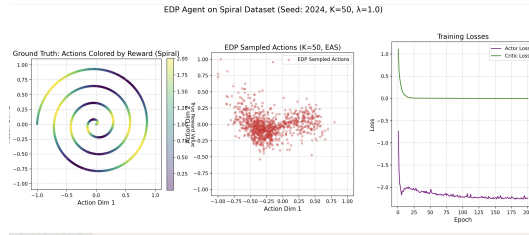


Figure 3: Mosquito coil shape

EDP demonstrated strong performance across various simple bandit tasks, encompassing scenarios such as Equal Gaussian Mixture, Increased Modal Complexity, and Mosquito Coil Shape distributions. Notably, this excellent performance by EDP is achieved after significantly overcoming two critical bottlenecks present in prior diffusion model-based reinforcement learning approaches: first, the high computational costs associated with full Markov chain propagation during training; and second, the incompatibility with likelihood-based RL algorithms due to the intractability of diffusion models' likelihood functions.